Chapter 5 — States and Phases of Matter

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 109–111

Q.1 — Multiple Choice Questions (with correct option and reason)

I. London dispersion forces are the only forces present among:

(a) molecules of water in liquid state

(b) atoms of helium in gaseous state at high temperature

(c) molecules of solid iodine

(d) molecules of hydrogen chloride gas

Answer: (c) molecules of solid iodine — Solid I_2 is non-polar, so only London forces act between I_2 molecules.

Helium gas (b) also has only London forces, so both (b) and (c) are physically correct; (c) is the usual intended molecular example.

II. When the vapour pressure of a liquid equals the external pressure, what phenomenon occurs?

(a) Sublimation

(b) Condensation

(c) Boiling

(d) Freezing

Answer: (c) Boiling — Boiling occurs when the vapour pressure equals the external pressure.

III. When water freezes at 0 °C, its density decreases due to:

(a) cubic structure of ice

(b) empty spaces present in the structure of ice

(c) decrease in volume

(d) decrease in viscosity

Answer: (b) empty spaces present in the structure of ice — Hydrogen bonding forms an open hexagonal lattice with empty spaces, increasing volume and lowering density.

IV. Which of the following is the correct sequence of increasing ΔHv values of the substances mentioned?

(a) $H_2O > NH_3 > F_2$

(b) $F_2 > NH_3 > H_2O$

(c) $NH_3 > H_2O > F_2$

(d) $H_2O > F_2 > NH_3$

Answer: (a) $H_2O > NH_3 > F_2$ — Strength of intermolecular forces: H_2O (strong H-bonding) > NH_3 (H-bonding) > F_2 (London only), so ΔHv follows $H_2O > NH_3 > F_2$.

The question says “increasing” but the options use “>” (decreasing). Option (a) gives the correct relative order; the increasing order is $F_2 < NH_3 < H_2O$.

V. Surface tension of a liquid is due to:

(a) inward pull of surface molecules

(b) upward pull from the surface

(c) collision of molecules

(d) repulsive forces

Answer: (a) inward pull of surface molecules — Surface molecules experience a net inward pull because they are not surrounded by liquid molecules on all sides.

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VI. Which change of state involves overcoming only London dispersion forces?

(a) Melting of ice ($H_2O(s) \rightarrow H_2O(l)$)

(b) Boiling of ethanol ($CH_3CH_2OH(l) \rightarrow CH_3CH_2OH(g)$)

(c) Sublimation of iodine ($I_2(s) \rightarrow I_2(g)$)

(d) Dissolving sodium chloride in water

Answer: (c) Sublimation of iodine ($I_2(s) \rightarrow I_2(g)$) — I_2 is non-polar; sublimation overcomes only London forces. Ice/ethanol involve H-bonds; NaCl dissolution involves ion–dipole forces.

VII. In which of the following substances are London dispersion forces the only significant intermolecular forces present?

(a) Ammonia (NH_3)

(b) Water (H_2O)

(c) Methane (CH_4)

(d) Hydrogen fluoride (HF)

Answer: (c) Methane (CH_4) — CH_4 is non-polar and has no H-bonding, so only London forces are significant.

VIII. Which of the following is a characteristic property of crystalline solids?

(a) They have a range of melting points.

(b) They are isotropic.

(c) They have a definite and sharp melting point.

(d) They lack a regular arrangement of particles.

Answer: (c) They have a definite and sharp melting point. — Crystalline solids have a regular lattice and melt sharply at a definite temperature.

IX. Which of the following liquids would you expect to have the highest viscosity at a given temperature?

(a) Water (H_2O)

(b) Ethanol (CH_3CH_2OH)

(c) Diethyl ether ($CH_3CH_2OCH_2CH_3$)

(d) Glycerol ($CH_2(OH)CH(OH)CH_2(OH)$)

Answer: (d) Glycerol ($CH_2(OH)CH(OH)CH_2(OH)$) — Glycerol has three –OH groups and extensive hydrogen bonding, giving the highest viscosity.

X. Which type of intermolecular force is present in all types of molecule regardless of their polarity?

(a) Dipole–dipole forces

(b) Hydrogen bonds

(c) London dispersion forces

(d) Ion–dipole forces

Answer: (c) London dispersion forces — London dispersion forces arise from instantaneous dipoles and exist in all atoms and molecules.

XI. Liquid crystals exhibit properties:

(a) Only like solids.

(b) Only like liquids.

(c) Between solids and liquids.

(d) Unlike solids or liquids.

Answer: (c) Between solids and liquids. — Liquid crystals have fluidity like liquids but some ordered arrangement like solids.

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Q.2 — Short Answer Questions

a. Explain, at the molecular level, why evaporation leads to a cooling effect.

The fastest (highest-energy) molecules escape from the liquid surface during evaporation. The average kinetic energy of the remaining molecules falls, so the temperature of the liquid decreases, producing cooling.

b. Why liquids with stronger intermolecular forces tend to have lower rates of evaporation at a given temperature?

Stronger intermolecular forces hold molecules more tightly in the liquid, so fewer molecules have enough energy to escape. Hence the evaporation rate is lower.

c. One feels a sense of cooling under the fan after a bath. Explain.

Water on the skin evaporates; the fan increases air flow and speeds evaporation. Evaporation absorbs latent heat from the skin, so the body feels cool.

d. How is a dynamic equilibrium established during evaporation of a liquid in a closed vessel at constant temperature?

Initially evaporation is faster than condensation. As vapour builds up, the condensation rate increases until it equals the evaporation rate. At this point the amounts of liquid and vapour remain constant, although both processes continue — a dynamic equilibrium.

e. The boiling point of water is different at Lahore and Murree hills.

Murree is at higher altitude where atmospheric pressure is lower than in Lahore. Water boils when its vapour pressure equals the external pressure, so it boils at a lower temperature on the hills and at a higher temperature in Lahore.

f. Discuss two significant consequences of the lower density of ice compared to liquid water in natural environments.

(i) Ice floats on water, insulating the water below so aquatic life survives in winter. (ii) Freezing of water in rocks/soil causes expansion and weathering (frost action), shaping landscapes.

g. Why does the boiling point of a liquid increase when the external pressure rises?

A higher external pressure requires a higher vapour pressure for boiling. Since vapour pressure increases with temperature, the liquid must be heated to a higher temperature to boil.

h. Mention four items in which liquid crystals are used.

Liquid-crystal displays (LCDs) in watches, calculators, laptop/TV screens, digital thermometers and temperature-sensing strips.

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i. How do you differentiate between crystalline solids and amorphous solids?

Property	Crystalline solid	Amorphous solid
Arrangement	Regular, repeating lattice	Irregular/random
Melting point	Sharp, definite	Melts over a range
Nature	Anisotropic	Isotropic
Cleavage	Cleaves along definite planes	Irregular breakage
Example	NaCl, quartz	Glass, rubber

j. Propanone, propanol and butane have similar molar masses. List them in order of increasing boiling points and explain.

Increasing boiling point: butane < propanone < propanol. Butane has only London forces; propanone has dipole–dipole forces; propanol has hydrogen bonding (strongest), so it boils highest.

k. How is hydrogen bonding responsible for the relatively high surface tension of water?

Water molecules at the surface are pulled inward by strong hydrogen bonds with neighbouring molecules. This strong inward cohesive force gives water a high surface tension.

l. What type of intermolecular forces will dominate in: (i) NH_3 (ii) Ar (iii) CH_3COCH (iv) CH_3OH?

(i) NH_3: hydrogen bonding. (ii) Ar: London dispersion forces. (iii) CH_3COCH_3 (propanone): dipole–dipole forces. (iv) CH_3OH: hydrogen bonding.

m. The boiling points and molar masses of CH_4, NH_3 and H_2O are tabulated. Suggest reasons for the differences in boiling points.

Although molar masses are similar (16, 17, 18), boiling points rise CH_4 (112 K) < NH_3 (240 K) < H_2O (373 K). CH_4 has only weak London forces. NH_3 forms hydrogen bonds. H_2O forms more extensive/stronger hydrogen bonds (two H and two lone pairs per molecule), so it has the highest boiling point.

Substance	Boiling point (K)	Molar mass (g mol^{-1})	Dominant force
CH_4	112	16	London dispersion
NH_3	240	17	Hydrogen bonding
H_2O	373	18	Extensive hydrogen bonding

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Descriptive / Long Questions

Q.3 What are London dispersion forces? Give examples, and discuss the factors affecting these forces.

London dispersion forces are weak intermolecular attractions caused by instantaneous (temporary) dipoles arising from momentary uneven distribution of electrons. These induce dipoles in neighbouring molecules, producing attraction. They exist in all atoms and molecules, including non-polar ones (He, CH_4, I_2).

Factors: (i) Number of electrons / molar mass — larger electron clouds are more polarisable, giving stronger forces (I_2 > Br_2 > Cl_2 > F_2). (ii) Molecular size/surface area — larger, elongated molecules have greater contact area and stronger forces than compact isomers. (iii) Shape — straight-chain molecules have higher boiling points than branched isomers.

Q.4 Hydrogen bonding is present in H_2O, NH_3, HF, $(CH_3)_2CO$ and $CHCl_3$ molecules. Sketch structures and discuss briefly.

A hydrogen bond forms when H bonded to a highly electronegative atom (F, O, N) is attracted to a lone pair on another electronegative atom.

H_2O: each molecule can form up to four H-bonds (two H donors, two lone-pair acceptors) → extensive network. **NH_3:** N–H···N hydrogen bonds (weaker than water because N is less electronegative and has only one lone pair). **HF:** F–H···F chains (strong H-bonds).

$(CH_3)_2CO$ (propanone): the carbonyl O accepts H-bonds from suitable donors (e.g. water or $CHCl_3$); it has no O–H to donate. **$CHCl_3$:** the C–H bond is polarised by three Cl atoms, so it can act as a weak H-bond donor to the oxygen of propanone (C–H···O).

Pure propanone cannot hydrogen-bond to itself (no H bonded to O/N/F); it participates as an acceptor with donors such as water or chloroform.

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Q.5 Discuss the structural changes when water turns into ice. Justify the empty spaces in its crystals as compared to H_2O at 4 °C and the lower density of ice.

In liquid water, hydrogen bonds are continually breaking and reforming, allowing molecules to pack relatively closely. On freezing, each water molecule forms four stable hydrogen bonds in a tetrahedral arrangement, producing an open hexagonal lattice with large empty spaces.

This open structure occupies more volume than the liquid, so ice is less dense than water (maximum density of water is at 4 °C). Hence ice floats and water expands on freezing.

Q.6 How do liquid crystals resemble liquids and solids? Give their uses in daily life.

Liquid crystals flow like liquids (they take the shape of the container) but retain some orientational/orderly arrangement like solids, giving anisotropic optical properties. They are therefore intermediate between crystalline solids and isotropic liquids.

Uses: LCD screens (watches, calculators, phones, TVs), digital thermometers, temperature-sensing strips, and optical devices.

Q.7 Describe the following properties of crystalline solids: (i) Geometrical shape (ii) Melting point (iii) Cleavage plane (iv) Habit of a crystal.

(i) Geometrical shape: crystalline solids have a definite, regular geometrical shape with flat faces and sharp edges due to ordered internal arrangement.

(ii) Melting point: they melt sharply at a definite temperature because the lattice breaks down uniformly.

(iii) Cleavage plane: they split along definite planes (cleavage planes) when cut, giving smooth surfaces.

(iv) Habit: the characteristic external shape in which a crystal grows (e.g. cubic NaCl, needle-like, prismatic), determined by internal structure and growth conditions.

Numerical Problems (fully worked)

Q.8 A gas sample has mass 0.560 g, volume 2.87×10^{-4} m³ at 300 K and pressure 1.01×10^{5} Pa. Calculate the molar mass. (R = 8.31 J K⁻¹ mol¹)

Given: m = 0.560 g; V = 2.87×10^{-4} m³; T = 300 K; P = 1.01×10^{5} Pa; R = 8.31 J K^{-1} mol^{-1}

Formula: PV = nRT; n = PV/RT; M = m/n = mRT/(PV)

n = $(1.01 \times 10^{5} \times 2.87 \times 10^{-4})/(8.31 \times 300)$ = 28.99/2493 = 0.01163 mol.

M = 0.560/0.01163 = 48.2 g mol^{-1}.

Answer: Molar mass ≈ 48.2 g mol^{-1}

Q.9 150 cm³ of a volatile liquid is completely vaporized at 98 °C and 1.01×10^{5} Pa. The vapour mass is 0.495 g. Determine the molecular mass. (R = 8.314 J K^{-1} mol^{-1})

Given: V = 150 cm³ = 1.50×10^{-4} m³; T = 98 + 273 = 371 K; P = 1.01×10^{5} Pa; m = 0.495 g

Formula: M = mRT/(PV)

n = PV/RT = $(1.01 \times 10^{5} \times 1.50 \times 10^{-4})/(8.314 \times 371)$ = 15.15/3084.5 = 0.004912 mol.

M = 0.495/0.004912 = 100.8 g mol^{-1}.

Answer: Molecular mass ≈ 101 g mol^{-1}

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Quick Check Solutions (in-chapter practice boxes)

Quick Check 5.1 (book p. 93)

a. Why gases can be compressed easily.

Gas particles are far apart with large empty spaces between them, so applying pressure can push them closer together.

b. What is the Joule–Thomson effect?

The cooling (or heating) of a real gas when it expands adiabatically through a porous plug/throttle from high to low pressure, due to work done against intermolecular attractions.

c. 84 g of a gas occupies 8 dm³ at −90 °C and 7 atm. Calculate relative molecular mass.

T = 183 K. M = mRT/PV = (84 × 0.0821 × 183)/(7 × 8) = 22.5 g mol^{-1}.

Quick Check 5.2 (book p. 96)

a. Why straight-chain hydrocarbons in gasoline are converted to branched-chain ones.

Branched hydrocarbons burn more smoothly and have higher octane ratings, reducing knocking in the engine.

b. Forces present among CCl_4 and SiF_4 molecules.

Both are non-polar (symmetrical tetrahedral), so only London dispersion forces act between their molecules.

c. Differentiate id–id and pd–pd forces with examples.

id–id (instantaneous dipole–induced dipole / London) forces occur in all molecules, e.g. CH_4, I_2. pd–pd (permanent dipole–permanent dipole) forces occur between polar molecules, e.g. HCl, CH_3Cl.

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Quick Check 5.3 (book p. 99)

a. Can CHF_3 make a hydrogen bond? Explain.

Not a classical hydrogen bond: the H is bonded to C, not to F/O/N, so it is not sufficiently δ+ to donate a strong H-bond. (The F atoms can accept H-bonds from other donors.)

b. Show a hydrogen bond between two ethanol molecules.

CH_3CH_2–O–H···O(H)–CH_2CH_3: the H of one –OH group is attracted to the lone pair on the O of another molecule.

c. Forces in $CH_3CH_2CH_3$, CH_3CH_2OH, CH_3CH_2Cl; arrange by increasing boiling point.

Propane: London only; chloroethane: dipole–dipole; ethanol: hydrogen bonding. Increasing b.p.: propane < chloroethane < ethanol.

d. Why the boiling-point difference between alkane and alcohol decreases as molecular size increases.

As the carbon chain grows, London forces increase and become a larger part of the total attraction, so the extra contribution from hydrogen bonding becomes relatively smaller.

e. Why ethanoic acid forms dimers in pure form but not in aqueous solution.

In pure acid, two molecules pair through two O–H···O hydrogen bonds, forming a dimer. In water, acid molecules hydrogen-bond with water instead, breaking the dimers.

Quick Check 5.4 (book p. 101)

a. Arrange acetone, ethanol and dimethyl ether by increasing surface tension.

Dimethyl ether < acetone < ethanol. Ethanol has hydrogen bonding (strongest cohesion), acetone has dipole–dipole, ether has weaker dipole/London forces.

b. Why CCl_4 has higher viscosity than $CHCl_3$ but less than ethanol.

CCl_4 is larger/more polarisable than $CHCl_3$, giving stronger London forces and higher viscosity. Ethanol has hydrogen bonding, giving still stronger cohesion and higher viscosity.

c. Why honey is more viscous than water.

Honey contains large sugar molecules with many –OH groups that form extensive hydrogen bonds, resisting flow much more than water.

d. Which is more viscous: glycerine or hexane? Why?

Glycerine, because its three –OH groups form extensive hydrogen bonds; hexane has only weak London forces.

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Quick Check 5.5 (book p. 102)

a. Which liquid has higher vapour pressure: (i) alcohol/glycerine (ii) petrol/kerosene (iii) mercury/water?

(i) Alcohol (glycerine has stronger H-bonding). (ii) Petrol (more volatile than kerosene). (iii) Water (mercury has very low vapour pressure).

b. Which is more viscous: glycerine or kerosene?

Glycerine (extensive hydrogen bonding).

c. Acetone feels colder than water on the hand. Explain.

Acetone has weaker intermolecular forces, so it evaporates faster and absorbs heat from the skin more quickly, producing a stronger cooling effect.

d. Why evaporation is faster at higher temperatures.

More molecules have enough kinetic energy to overcome intermolecular forces and escape from the surface.

e. Why we feel cool near a river bank.

Evaporation of water absorbs latent heat from the surroundings, lowering the temperature of the nearby air.

Quick Check 5.6 (book p. 104)

a. Why cooking is difficult at high altitudes.

Atmospheric pressure is lower, so water boils below 100 °C; the lower temperature cooks food more slowly.

b. Why food cooks faster in a pressure cooker.

The sealed cooker raises the pressure, raising the boiling point of water above 100 °C, so food cooks at a higher temperature.

c. Why water (100 °C) has a higher boiling point than ethanol (78 °C) although both have H-bonds.

Water forms more extensive hydrogen bonds (two H atoms and two lone pairs per molecule) than ethanol, so more energy is needed to separate its molecules.

Quick Check 5.7 (book p. 107)

a. Why solids have very low compressibility and expansion.

Particles are tightly packed in fixed positions with strong forces and little empty space, so they resist compression and expand only slightly on heating.

b. What is meant by the habit of a crystal?

The characteristic external shape in which a crystal grows (e.g. cubic, needle-like, prismatic), determined by its internal structure and growth conditions.

c. Why solids do not undergo translatory motion.

Particles are held in fixed lattice positions and can only vibrate about those positions; they cannot move freely from place to place.

Quick Check 5.8 (book p. 109)

a. Properties of liquid crystals in which they resemble solids.

They have some ordered/orientational arrangement and anisotropic optical properties, like crystalline solids.

b. Properties in which they resemble liquids.

They flow and take the shape of the container, like liquids.

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c. Which property makes them useful in temperature-sensing devices?

Their optical properties (colour/pitch of reflected light) change sensitively with temperature.

Quick Check 5.4(d): hexane is printed as C_6H_6 (benzene); the intended alkane is C_6H_{14}.

Textbook checks flagged in this chapter

• p. 109, MCQ I: both helium gas (b) and solid iodine (c) have only London forces; the question has two physically correct options.

• p. 109, MCQ IV: the word “increasing” conflicts with the “>” notation in the options; the correct relative order is $H_2O > NH_3 > F_2$.

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